Linear Regression Assignment

Energy Consumption Prediction

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1. Overview

This assignment covers predicting building energy consumption using simple and multiple linear regression, built from scratch using gradient descent. The dataset has four input features: Square Footage, Number of Occupants, Appliances Used, and Average Temperature. Six models were trained: one per feature for simple regression, plus two multiple regression trials with different feature sets.

2. Setup

Parameter	Value
Normalization	Min-Max Scaling to [0, 1]
Weight Init	m = 0, c = 0
Learning Rates Tried	0.001 | 0.01 | 0.04 | 0.06 | 0.08 | 0.1 | 0.11 | 0.15 | 0.17 | 0.2
Epochs Tried	30 | 50 | 60 | 100 | 200 | 300
Best LR	0.1
Best Epochs	1000
Error Metric	MSE & RMSE

3. MSE Results

Final MSE and RMSE for each model at lr=0.1, epochs=1000:

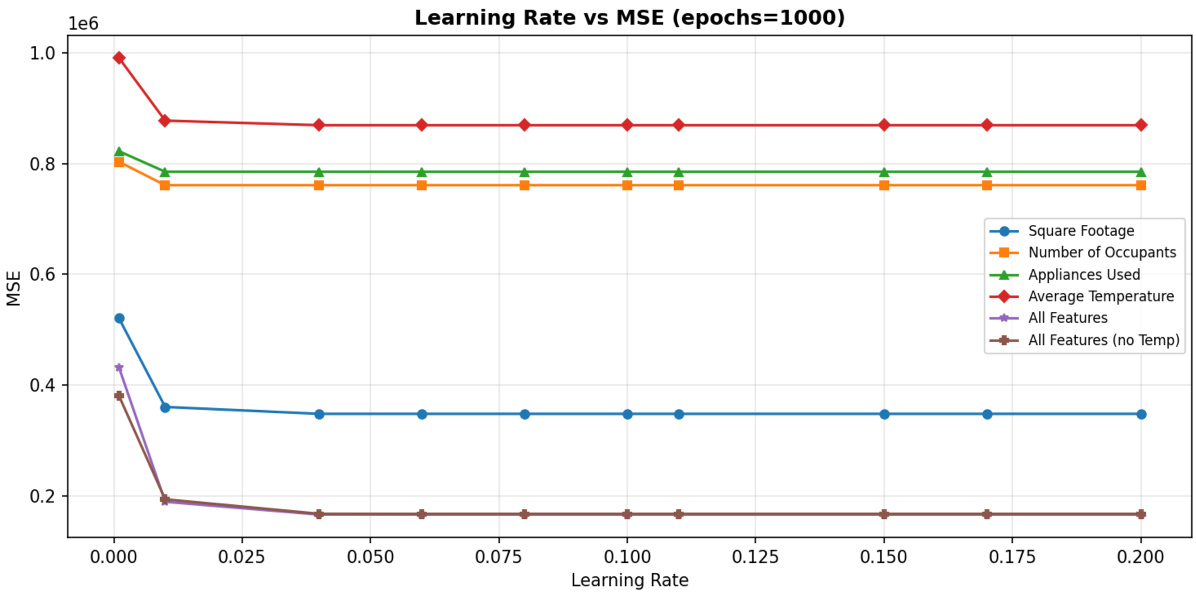
Model	Type	MSE	RMSE
Square Footage	Simple LR	347,708.84	589.67
Number of Occupants	Simple LR	760,853.12	872.27
Appliances Used	Simple LR	785,062.79	886.04
Average Temperature	Simple LR	869,167.22	932.29
All Features	**Multiple LR**	**165,639.57**	**406.99**

All Features (no Temp)	**Multiple LR**	**167,048.80**	**408.72**

4. Plots

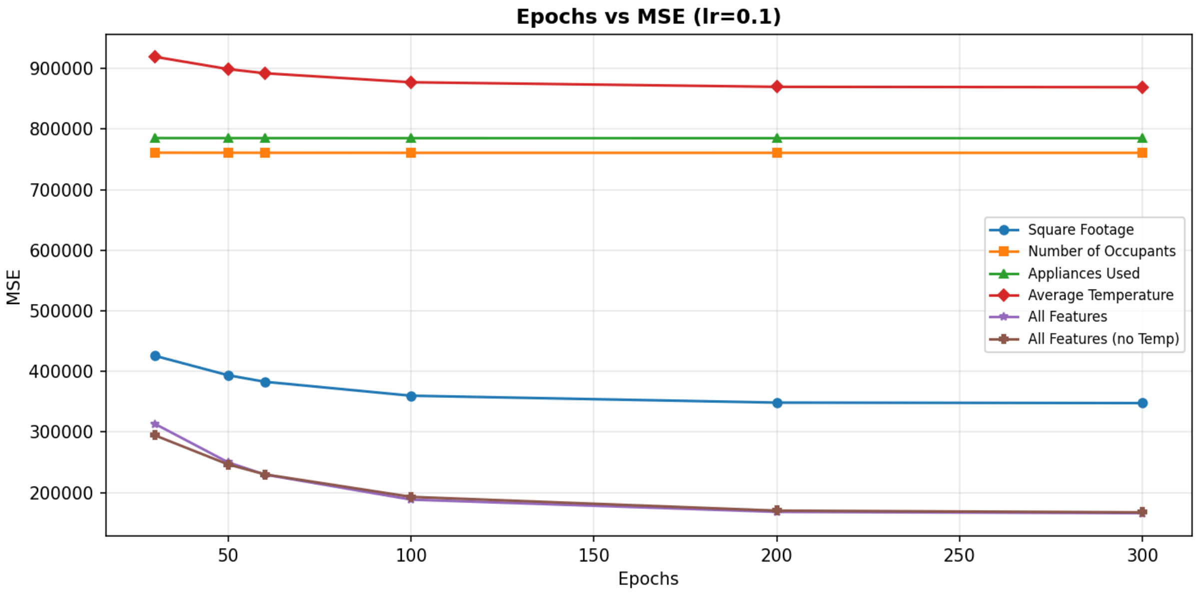
Learning Rate vs MSE

MSE drops sharply from lr=0.001 to lr=0.04, then stabilizes. lr=0.1 was picked as the best since going beyond it changes nothing.



Epochs vs MSE

All models improve with more epochs. The multiple regression models drop more noticeably in the early stages. Single-feature models tend to flatten out faster since there is less to learn.



5. Conclusion

Best single feature: Square Footage came out on top with MSE ~347,708 and RMSE ~589. It has the clearest linear relationship with energy consumption out of the four features.

Worst single feature: Average Temperature had the highest MSE (~869,167). On its own it is a poor predictor, though it still contributes a bit when combined with others.

Multiple regression — all features: Using all four features brought MSE down to ~165,639, which is roughly half the best single-feature result. This is the strongest model overall.

Multiple regression — without temperature: Removing Average Temperature raised the MSE slightly to ~167,048. The difference is small, but keeping it in is still marginally better.

Learning rate: lr=0.1 worked well. Below 0.01 the model was still converging slowly at 1000 epochs, and anything above 0.1 gave no improvement.

Epochs: Models were mostly converged by 300 epochs at lr=0.1. Pushing to 1000 gave a small extra gain but nothing dramatic.